

OAKPOINT, MB, Canada

WMO: 711440

Lat: 50.500N

Lon: 98.037W

Elev: 250

StdP: 98.36

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.4	(c) -29.1	(d) -34.7	(e) 0.1	(f) -31.3	(g) -32.3	(h) 0.2	(i) -29.0	(j) 13.3	(k) -15.1	(l) 12.0	(m) -15.1	(n) 2.9	(o) 350	(p) 0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	27.4	21.9	26.0	20.9	24.7	20.1	23.4	26.4	22.2	25.1	21.0	24.0	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.6	25.7	21.1	16.2	24.5	19.9	15.0	23.3	71.4	26.5	66.5	25.1	62.2	24.0	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.6	11.0	9.7	-34.8	30.3	3.0	1.6	-36.9	31.5	-38.7	32.4	-40.3	33.4	-42.5	34.6		
			-34.6	25.1	3.0	1.4	-36.8	26.1	-38.5	26.8	-40.2	27.6	-42.4	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-16.8	-15.0	-7.7	1.6	9.6	16.6	19.5	18.4	13.1	5.2	-4.2	-12.8
	DBStd	14.02	7.71	7.58	7.36	5.89	5.02	3.79	3.14	3.31	4.31	4.42	6.09	7.11
	HDD10.0	3718	829	699	549	257	71	2	0	0	18	159	427	706
	HDD18.3	5957	1088	932	808	503	273	76	24	41	163	407	677	964
	CDD10.0	939	0	0	0	4	57	199	296	261	112	10	0	0
	CDD18.3	136	0	0	0	0	2	23	61	43	7	0	0	0
	CDH23.3	622	0	0	0	0	8	94	291	209	21	0	0	0
	CDH26.7	77	0	0	0	0	1	11	34	29	2	0	0	0
Wind	WSAvg	4.6	4.5	4.4	4.5	4.4	4.5	4.4	4.5	4.6	4.8	5.3	5.1	4.6
Precipitation	PrecAvg	517	21	17	29	34	65	89	68	56	47	39	28	29
	PrecMax	712	56	43	66	76	145	156	121	128	105	88	118	50
	PrecMin	400	8	0	6	10	20	30	21	16	18	6	2	7
	PrecStd	75	11	12	18	17	36	34	28	30	22	22	26	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	2.7	7.5	18.2	24.3	28.1	29.4	29.5	25.6	18.3	9.8	2.0
		MCWB	0.8	1.7	4.9	10.3	15.8	22.3	23.9	22.7	19.5	13.6	7.4	0.9
	2%	DB	-0.9	0.9	4.2	14.6	21.4	25.7	27.6	27.2	23.2	15.5	7.0	-0.1
		MCWB	-1.4	-0.1	2.4	9.1	14.8	19.8	22.5	21.2	18.4	12.1	5.2	-0.7
	5%	DB	-3.1	-0.7	2.7	11.8	19.0	24.2	26.1	25.6	21.4	13.3	5.0	-1.5
		MCWB	-3.5	-1.5	1.6	7.2	13.6	19.3	21.7	20.3	17.4	10.5	3.7	-2.0
	10%	DB	-5.2	-3.5	1.5	9.4	17.0	22.7	24.9	24.2	19.7	11.6	3.4	-3.0
		MCWB	-5.6	-4.1	0.6	6.2	12.5	18.4	20.9	19.5	16.3	9.3	2.3	-3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.9	5.3	11.5	18.2	24.1	25.6	24.8	20.9	14.7	7.7	1.1
		MCDB	1.4	2.7	7.3	16.6	21.9	27.0	28.2	28.2	23.9	17.8	9.3	1.8
	2%	WB	-1.2	0.1	2.8	9.6	16.1	21.6	23.7	22.9	19.3	12.4	5.3	-0.5
		MCDB	-0.8	1.0	3.9	13.4	19.9	24.4	26.5	26.2	22.2	14.9	6.6	0.0
	5%	WB	-3.3	-1.4	1.7	8.1	14.7	20.2	22.6	21.6	18.1	11.0	3.8	-1.8
		MCDB	-2.9	-0.5	2.6	11.5	18.0	23.0	25.5	24.7	20.9	13.0	4.9	-1.4
	10%	WB	-5.5	-3.9	0.8	6.2	13.3	18.9	21.5	20.4	16.8	9.4	2.3	-3.2
		MCDB	-5.0	-3.4	1.5	9.5	16.4	22.0	24.4	23.4	19.5	11.5	3.3	-2.8
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.3	9.4	9.2	9.4	8.5	8.3	9.0	8.3	6.7	6.0	7.3
		MCDBR	8.9	7.5	7.9	12.3	11.8	9.9	8.9	10.3	10.4	9.4	6.5	7.3
		MCWBR	8.7	6.9	6.5	7.5	7.0	6.4	6.1	6.6	6.8	6.4	5.1	7.0
	5% WB	MCDBR	8.7	7.2	7.6	11.9	9.9	8.8	8.5	9.4	9.4	8.7	6.1	7.2
		MCWBR	8.5	6.6	6.5	7.5	6.6	6.4	6.4	6.7	6.8	6.4	4.9	7.0
Clear-Sky Solar Irradiance	taub		0.253	0.282	0.306	0.355	0.371	0.379	0.392	0.379	0.346	0.300	0.265	0.242
	taud		2.268	2.202	2.291	2.342	2.368	2.381	2.373	2.407	2.482	2.561	2.422	2.295
	Ebn at Noon		789	850	898	887	885	878	859	854	850	840	781	751
	Edh at Noon		67	95	105	113	115	115	114	105	88	67	58	56
All-Sky Solar Radiation	RadAvg		1.30	2.50	3.84	5.08	5.50	5.83	6.13	5.09	3.64	2.13	1.15	0.93
	RadStd		0.12	0.17	0.29	0.35	0.49	0.43	0.27	0.38	0.30	0.28	0.16	0.09

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (24 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) -0.57	(i) -0.79	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page